



DW_lp_multifunc

Low Power Fixed-Point Multi-Function Unit

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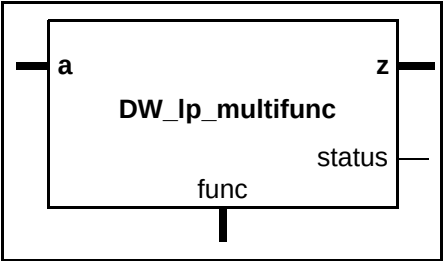
Features and Benefits

Revision History

- Parameterized precision of operands and number of functions to be implemented.
- It implements any combination of the following functions:

$$\frac{1}{x}, \frac{1}{\sqrt{x}}, \sqrt{x}, \sin(\pi x), \cos(\pi x), \log_2(x), 2^x$$

- One function is computed at a time
- All functions produce monotonic results (depends on input range--see [Table 1-9](#) on page 5)
- Shared polynomial approximation unit provides a solution with reduced area.



Note

You must use VCS to simulate the DW_lp_multifunc component. Non-VCS simulators are not supported.

Description

DW_lp_multifunc is a fixed-point multi-function unit that implements any combination of seven functions: reciprocal, square root, reciprocal square root, sine, cosine, base-2 logarithm, and base-2 exponential. The particular set of functions to be implemented is determined by the *func_select* parameter, where each bit of the parameter value corresponds to a function. At any given time, the unit computes one of the implemented functions as determined by input *func*, which must be a "valid one-hot code". A valid one-hot code is one that has only one bit set to 1, and this bit corresponds to one of the bits set to 1 in the *func_select* parameter value. (For details, see [Table 1-5](#) on page 2.) If input *func* is not a valid one-hot code, the *status* flag is set and output *z* should be ignored.

The *op_width* parameter determines the bit-width of input and output values.

Table 1-1 Pin Descriptions

Pin Name	Width	Direction	Function
a	(<i>op_width</i> + 1) bits	Input	Input data
func	16 bit	Input	Function selection; details are shown in Table 1-5 on page 2
z	(<i>op_width</i> + 2) bits	Output	Output data
status	1 bit	Output	Flag of invalid operation

Table 1-2 Parameter Description

Parameter	Values	Description
op_width	3 to 24 Default: 24	Word length of input and output
func_select	1 to 127 Default: 127	Determines which of the supported functions are implemented (one bit for each function); details are shown in Table 1-5

Table 1-3 Synthesis Implementations

Implementation Name	Implementation	License Feature Required
rtl	Low Power Synthesis model	- DesignWare (P-2019.03 and later) - DesignWare-LP^a (before P-2019.03)

- a. For versions before P-2019.03, you must enable minPower as follows:
`set_synthetic_library {dw_foundation.sldb dw_minpower.sldb}`

Table 1-4 Simulation Models

Model	Function
dw/sim_ver/DW_lp_multifunc.v	Verilog simulation model ^a source code You must use VCS to simulate the DW_lp_multifunc component. Non-VCS simulators are not supported.

- a. To use this simulation model, the '+v2k' options needs to be used on the VCS command line.

Parameter *func_select* and Input *func*

The *func_select* parameter specifies the set of functions to be implemented. The valid range of the *func_select* parameter is from 1 to 127. [Table 1-5](#) presents the *func_select* parameter values that implement the various supported functions, along with the one-hot codes for the *func* input port that select each function during usage.

Table 1-5 *func_select* Parameter Implementation (design) and *func* Function Selection (usage)

func_select Bit	Function Description	func 16-bit select value
2 ⁰	reciprocal, 1/a	0000_0000_0000_0001
2 ¹	square root of a	0000_0000_0000_0010
2 ²	reciprocal square root of a	0000_0000_0000_0100
2 ³	sine, sin(π a)	0000_0000_0000_1000
2 ⁴	cosine, cos(π a)	0000_0000_0001_0000

Table 1-5 **func_select** Parameter Implementation (design) and func Function Selection (usage) (Continued)

func_select Bit	Function Description	func 16-bit select value
2^5	base-2 logarithm, $\log_2 a$	0000_0000_0010_0000
2^6	base-2 exponential, 2^a	0000_0000_0100_0000
$2^7 - 2^{15}$		Reserved for future support

As shown in [Table 1-5](#) on page 2, when only the reciprocal function is required, the *func_select* parameter is set to 1 (16'h0001).

When the reciprocal, reciprocal square root, and base-2 logarithm functions are needed, the *func_select* parameter is set to 37 (16'h0025).

To implement all seven functions, the *func_select* parameter is set to 127 (16'h007f).

During operation, the input port *func* specifies the single function that is to be computed at that time. The *func* port is a 16-bit input port, and it must be set to a "valid one-hot code". A valid one-hot code has only one bit set to 1, and this bit corresponds to one of the bits set to 1 in the *func_select* parameter value. The one-hot codes for all valid functions are shown in [Table 1-5](#) on page 2.

If *func* receives a code for a function that was not implemented by the *func_select* parameter, the *status* flag is set. If *func* does not receive a valid one-hot encoded value, the *status* flag is set. An exceptional case occurs when only one function is implemented (*func_select* has a value of either 1, 2, 4, 8, 16, 32 or 64). In this case, the *status* output is not set to 1 for any of the one-hot codes shown in [Table 1-5](#) on page 2, and the *z* output is generated based on the single function that the component implements. Note that the *status* output is set to 1 if the *func* input is not one of the one-hot codes in [Table 1-5](#) on page 2.

Input and Output Ranges of Functions

Valid input ranges of the seven functions are described in [Table 1-6](#). If the input value is out of range, the *status* flag is set to 1, indicating an incorrect output value.

Table 1-6 **Input and Output Ranges**

func_select Bit	Function Description	Input Range	Output Range
2^0	reciprocal, $1/a$	$1 \leq a < 2$	$0.5 < z \leq 1$
2^1	square root of a	$0.25 \leq a < 1$	$0.5 \leq z < 1$
2^2	reciprocal square root of a	$0.25 \leq a < 1$	$1 < z \leq 2$
2^3	sine, $\sin(\pi a)$	$0 \leq a < 2$	$-1 \leq z \leq 1$
2^4	cosine, $\cos(\pi a)$	$0 \leq a < 2$	$-1 \leq z \leq 1$
2^5	base-2 logarithm, $\log_2 a$	$1 \leq a < 2$	$0 \leq z < 1$
2^6	base-2 exponential, 2^a	$0 \leq a < 1$	$1 \leq z < 2$

Input and Output Format of Functions

Because valid input and output ranges are different among the seven functions, the number formats for the input and output ports vary. Therefore, the input format is one bit wider than *op_width* and the output format is two bits wider. The input format consists of one-bit integer part and *op_width*-bit fractional part. The output format has a two-bit integer part and *op_width*-bit fractional part. For example, [Table 1-7](#) shows the input and output formats with *op_width* = 24. For sine and cosine functions, the output value is represented in two's complement to express signed numbers.

Table 1-7 Input and Output Ranges

func_select	Function Description	Input Format	Output Format
2^0	reciprocal, $1/a$	$1.a_{23}a_{22} \dots a_10$	$0.z_{24}.z_{23}z_{22} \dots z_1z_0$
2^1	square root of a	$0.a_{23}a_{22} \dots a_1a_0$	$00.1z_{22} \dots z_1z_0$
2^2	reciprocal square root of a	$0.a_{23}a_{22} \dots a_1a_0$	$z_{25}z_{24}.z_{23}z_{22} \dots z_10$
2^3	sine, $\sin(\pi a)$	$a_{24}.a_{23}a_{22} \dots a_10$	$z_{25}z_{24}.z_{23}z_{22} \dots z_10$
2^4	cosine, $\cos(\pi a)$	$a_{24}.a_{23}a_{22} \dots a_10$	$z_{25}z_{24}.z_{23}z_{22} \dots z_10$
2^5	base-2 logarithm, $\log_2 a$	$1.a_{23}a_{22} \dots a_10$	$00.z_{23}z_{22} \dots z_1z_0$
2^6	base-2 exponential, 2^a	$0.a_{23}a_{22} \dots a_1a_0$	$01.z_{23}z_{22} \dots z_10$

Calculation Examples

[Table 1-8](#) shows the examples of the calculation results with *op_width* = 8 and *func_select* = 127. Numbers for input and output in [Table 1-8](#) are binary numbers.

Table 1-8 Calculation Examples (*op_width* = 8, *func_select* = 127)

func	Expression	Input	Output
16'h0001	$1/1.0111111 = 0.10101011$	1_0111_1110	00_1010_1011
16'h0002	$\text{SQRT}(0.11011101) = 0.11101101$	0_1101_1101	00_1110_1101
16'h0004	$1/\text{SQRT}(0.11011101) = 1.0100011$	0_1001_1101	01_0100_0110
16'h0008	$\sin(1.1101011\pi) = -0.011111$	1_1101_0110	11_1000_0010
16'h0010	$\cos(1.1101011\pi) = 0.110111$	1_1101_0110	00_1101_1110
16'h0020	$\log_2(1.1000101) = 0.10011111$	1_1000_1010	00_1001_1111
16'h0040	$2^{0.11100111} = 1.1101111$	1_1100_1110	01_1101_1110

Error Range and Monotonicity

The calculated output value of all functions guarantees 1-ulp error range. Each function has a different ulp position depending on the valid input range. The reciprocal, sine, cosine and base-2 logarithm functions have the input range larger than 1, and the ulp position has weight $2^{-(op_width-1)}$. The ulp position of square root, inverse square root and base-2 exponential has weight 2^{-op_width} .

The DW_lp_multifunc component produces monotonic results. The input ranges under which monotonic results are produced are shown in [Table 1-9](#).

Table 1-9 Input Range for Monotonic Results

	1/x	sqrt(x)	1/sqrt(x)	sin(x)	cos(x)	log2(x)	exp2(x)
<i>op_width</i> Input Range	All values of <i>op_width</i>	All values of <i>op_width</i>	All values of <i>op_width</i>	<i>op_width</i> <= 17	<i>op_width</i> <= 17	All values of <i>op_width</i>	All values of <i>op_width</i>

Verilog Description Example

If all functions except sine and cosine functions are implemented with *func_select* = 103 (7'b1100111) and the output needs to produce the result of the “square root” operation, the Verilog expressions needs to be described as follows.

```
DW_lp_multifunc #(8, 103) U1 (
    .a(9'b011011101),
    .func(16'h0002),
    .z(z),
    .status(status)
);
```

In this case, *z* becomes 10'b0011101101, as shown in [Table 1-8](#) on page 4, and *status* becomes 0. However, if the above module tries to calculate the sine function with *func* = 16'h0008, it signals the *status* flag, meaning that it is an invalid operation because DW_lp_multifunc does not have the netlist for the sine function with *func_select* = 103 (7'b1100111).

Single Function Implementation from DW_lp_multifunc

By selecting the *func_select* parameter to choose a single function, DW_lp_multifunc generates the netlist for the single function that has the similar functionality with single-function DW components, for example, DW_sqrt, DW_invsqrt, DW_sincos, DW_log2 and DW_exp2. The difference between the DW_lp_multifunc and these single-function components is the value of 1 ulp, due to the different implementation of polynomial approximation, but all evaluated results keep the error bound less than 1 ulp. Performance and area of the synthesis results are different depending on the synthesis constraints, library cells, and synthesis environments, for example. Therefore, by comparing performance and area between the single-function implementation and the DW_lp_multifunc in single-function mode, the DW_lp_multifunc provides more choices for the better synthesis results.

The following examples generate single functions.

Fixed-point Reciprocal (U1)

```
DW_lp_multifunc #(op_width, 1) U1 (  
    .a(a),  
    .func(16'h0001),  
    .z(z),  
    .status(status)  
);
```

Fixed-point square root (U2)

Results will be same as DW_sqrt except 1 ulp value.

```
DW_lp_multifunc #(op_width, 2) U2 (  
    .a(a),  
    .func(16'h0002),  
    .z(z),  
    .status(status)  
);
```

Fixed-point reciprocal square root (U3)

Results will be same as DW_invsqrt except 1 ulp value.

```
DW_lp_multifunc #(op_width, 4) U3 (  
    .a(a),  
    .func(16'h0004),  
    .z(z),  
    .status(status)  
);
```

Fixed-point sine and cosine (U4)

Results will be same as DW_sincos except 1 ulp value. Input func must be one of two values: 16'h0008 for sine and 16'h0010 for cosine evaluation.

```
DW_lp_multifunc #(op_width, 24) U4 (  
    .a(a),  
    .func(func),  
    .z(z),  
    .status(status)  
);
```

Fixed-point base-2 logarithm (U5)

Results will be same as DW_log2 except 1 ulp value.

```
DW_lp_multifunc #(op_width, 32) U5 (  
    .a(a),  
    .func(16'h0020),  
    .z(z),  
    .status(status)  
);
```

Fixed-point base-2 exponential (U6)

Results will be same as DW_exp2 except 1 ulp value.

```
DW_lp_multifunc #(op_width, 64) U6 (  
    .a(a),  
    .func(16'h0040),  
    .z(z),  
    .status(status)  
);
```

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Blocks User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_lp_multifunc_inst is
    generic (
        inst_op_width : POSITIVE := 24;
        inst_func_select : POSITIVE := 127
    );
    port (
        inst_a : in std_logic_vector(inst_op_width downto 0);
        inst_func : in std_logic_vector(15 downto 0);
        z_inst : out std_logic_vector(inst_op_width+1 downto 0);
        status_inst : out std_logic
    );
end DW_lp_multifunc_inst;

architecture inst of DW_lp_multifunc_inst is

begin

    -- Instance of DW_lp_multifunc
    U1 : DW_lp_multifunc
    generic map (
        op_width => inst_op_width,
        func_select => inst_func_select
    )
    port map (
        a => inst_a,
        func => inst_func,
        z => z_inst,
        status => status_inst
    );

end inst;
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_lp_multifunc_inst( inst_a, inst_func, z_inst, status_inst );

parameter inst_op_width = 24;
parameter inst_func_select = 127;

input [inst_op_width : 0] inst_a;
input [15 : 0] inst_func;
output [inst_op_width+1 : 0] z_inst;
output status_inst;

    // Instance of DW_lp_multifunc
    DW_lp_multifunc #(inst_op_width, inst_func_select) U1 (
        .a(inst_a),
        .func(inst_func),
        .z(z_inst),
        .status(status_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
June 2024	DWBB_202312.4	- Clarified the behavior of the <i>func_select</i> parameter and <i>func</i> port in “Description” on page 1, Table 1-2 on page 2, and “Parameter <i>func_select</i> and Input <i>func</i>” on page 2 - In “Parameter <i>func_select</i> and Input <i>func</i>” on page 2, corrected the description when the <i>func_select</i> parameter is set to implement only one of the available functions
December 2023	DWBB_202312.0	Updated date and version; no technical changes
January 2020	DWBB_201912.1	Corrected port names to be lowercase in Table 1-1 on page 1, examples, and format presentations
March 2019	DWBB_201903.0	Clarified some information about minPower in Table 1-3 on page 2
July 2018	DWBB_201806.1	- For STAR 9001366625, added a note about simulator support on page 1 and in Table 1-4 on page 2 - Added this Revision History table and the document links on this page

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